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Dayuan Wang

Bioinformatics/Biostatistics/Statistics

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SKILLS

Statistical skills	Causal inference, Generalized linear models, Mixed-effects models, Sample size / power analysis, Missing data methods, Hypothesis testing, Clinical trial design, Bayesian analysis, Survival analysis, Longitudinal analysis, Multiplicity adjustment, Semiparametric & nonparametric models, Drug discovery
Machine learning skills	Deep learning, Neural networks, Optimization, Penalized regression, Dimension reduction, Clustering
Omics & Bioinformatics	NGS, scRNA-seq, DNA-seq, Sequencing data pipelines, Multi-omics integration, Variant calling, Spatial-omics analysis, Copy number variation (CNV) detection, Single-cell trajectory analysis, Cell-cell communication analysis, Drug-target inference, Oncology, Immunology, Biomarker analysis
Tools and languages	R, Python, SAS, SQL, TensorFlow, Linux, HPC, GitHub, Markdown, $\text{\LaTeX}$

EDUCATION

PhD in Biostatistics	University of Florida, FL	Aug. 2022 – Aug. 2026 (Expected)
M.S. in Biostatistics	Rutgers University, NJ	Aug. 2018 – Dec. 2020
B.S. in Biology	Peking University, China	Sep. 2014 – July 2018

WORKING EXPERIENCE

Incoming Oncology Translational Bioinformatics Intern <i>Bristol Myers Squibb</i>	June 2026 – Aug. 2026 Cambridge, MA
- Apply deep learning and clinical trial data analysis to AI-powered multi-modal biomarker discovery for oncology treatment response. - Integrate peripheral biomarkers and clinical variables to develop predictive models and interpretable biomarker signatures. - Conduct reproducible analysis, model validation, performance benchmarking, and visualization to support cross-functional translational decision-making.	
Doctorate Research Assistant <i>Sepsis and Critical Illness Research Center, University of Florida</i>	Aug. 2022 – Current Gainesville, FL
- Drive translational research by applying multi-omics data to uncover immune mechanisms in critical illness, informing potential therapeutic targets. - Collaborated with clinicians, immunologists, and biomedical scientists to interpret results for biomedical studies. - Design and validate robust, reproducible end-to-end bioinformatics pipelines for sequencing data (e.g., scRNA-seq, bulk RNA-seq, microarray) using R/Python in HPC environment. - Identify immune biomarkers, key genes via multi-omics statistical modeling, supporting mechanism-of-action studies and hypothesis generation for clinical translation.	
Health Data Specialist <i>New Jersey Department of Health</i>	June 2019 – Feb. 2020 Trenton, NJ
- Applied real-world data analytics and generalized linear modeling (GLMs) to evaluate Hepatitis C burden and associated risk factors, supporting population health strategy and surveillance. - Developed geological risk prediction models to generate vulnerability heatmaps for Hepatitis C and HIV, enabling precision public health resource allocation. - Produced high-frequency COVID-19 surveillance reports and dashboards to support rapid-response public health decisions during the early pandemic.	

HONORS & REWARDS

PhD Fellowship in Artificial Intelligence	College of Public Health and Health Professions, University of Florida, FL	2024 – 2025
Department Achievement Award, Preliminary Exam Award	Department of Biostatistics, University of Florida, FL	2023
Student Research Travel scholarship	Department of Biostatistics, Rutgers University, NJ	2019
Outstanding Undergraduate Student Scholarship	Peking University, China	2014 – 2018
Gold Medal, 25th International Biology Olympiad	Bali, Indonesia	2014

RESEARCH PROJECTS

Deep Learning Genetic Biomarker Discovery: CNV Detection from Exome Data	Aug. 2023 - May 2025
Methodological Innovation: Developed CN-RNN, a hybrid Recurrent Neural Network model that accurately detect Copy Number Variants (CNV) from exome sequencing data, mitigating biases and noise.	

- **Quantified Superiority:** Outperformed existing tools on large-scale autism and congenital heart disease cohorts, improving the predictive performance (F1 score) by 20% while significantly reducing run time.
- Prepare CN-RNN as an open-source tool on GitHub for genetic biomarker discovery.

Statistical Learning Algorithm for Copy Number Alterations Detection in Spatial Multi-omics Data Aug. 2024 - Current

- **Pioneering Methodology:** Developing a first-of-its-kind statistical method to integrate spatial-DNA seq data for Copy Number Alteration (CNA) detection and cancer subclone classification, addressing challenges of data sparsity and noise.
- Enabling identification of spatially resolved genomic biomarkers and tumor heterogeneity patterns relevant to precision oncology and targeted therapy development.
- Preparing an open-source R package on GitHub to facilitate CNA analysis in spatial multi-omics datasets for cancer genomics research.

Sepsis Multi-Omics Immune Profiling (Multiple Studies) May 2023 - Current

- **Computational Pipeline Development:** Established single-cell multi-omics pipelines to profile innate and adaptive immune responses in murine and human sepsis.
- **Cell type Identification:** First to define the transcriptional and functional profiles of myeloid-derived suppressor cell (MDSC) subtypes in sepsis.
- **Pre-Clinical Drug Mechanism:** Demonstrated a statistically significant protective effect of BCG vaccination against sepsis in neonatal models, supported by evidence of metabolic and transcriptional reprogramming in myeloid cells (mechanism-of-action study).
- **Precision Medicine:** Described immune profile heterogeneity due to demographic factors (e.g., age and sex) to define age-specific distinct sepsis vulnerabilities for targeted therapeutic development.

PUBLICATIONS

D Wang, C Rodhouse, H Tang, et al. "Age but not sex modifies lymphoid immune responses in murine sepsis." (*Under Review*). (2026).

D Wang, F Qin, W Bao, R Bacher, D Chung, Q Lu, PA Efron, G Cai, F Xiao. "CN-RNN: a Deep Learning Framework for Copy Number Variation Detection with Exome Sequencing Data." *bioRxiv*, doi: 10.64898/2026.05.13.724920. (2026).

JC Rincon, **D Wang**, VE Polcz, et al. "Innate immune training in the neonatal response to sepsis." *Molecular Medicine*, 31(1), 159. (2025).

C Rodhouse, **D Wang**, H Tang, et al. "Age-and Sex-Driven Transcriptional and Metabolic Diversity in Myeloid-Derived Suppressor Cells After Mouse Sepsis." *bioRxiv*, 2025.10.06.680736. (2025).

JC Rincon, **D Wang**, VE Polcz, et al. "Single-cell transcriptome profiling reveals myeloid-derived suppressor cell (MDSC) heterogeneity in murine neonatal sepsis." *SHOCK*, 62(1), 16-17. (2024).

JA Balch, **D Wang**, R Wu, et al. "Transcriptomic evidence that critical illness induces an early expansion of multiple myeloid immune suppressive subsets." *SHOCK*, 62(1), 1-2. (2024).

CONFERENCE PRESENTATIONS

Joint Statistical Meetings (JSM)

Speaker: *CN-RNN: a Supervised Learning Framework for Copy Number Variation Detection with Sequencing Data* Aug. 2025
Nashville, TN

SHOCK Conference

Poster Presenter: *Sc-RNA Seq Identifies Age as Effect Modifier of Lymphocytes' Transcriptomic Changes in Murine Sepsis* June 2025
Boston, MA

SHOCK Conference

Poster Presenter: *Single-cell Transcriptome Profiling Reveals MDSC Heterogeneity in Murine Neonatal Sepsis* June 2024
Palm Beach, FL

Eastern North American Region of the Biometric Society (ENAR)

Poster Presenter: *An Integrative Deep Learning Model for Accurate Copy Number Variation Detection with NGS Data* Mar. 2024
Baltimore, MD

Northeast Epidemiology Conference and the Annual Infectious Disease Conference

Poster Presenter: *Distribution of Hepatitis C Virus Genotype and Associated Risk Factors in New Jersey, 2015 - 2018* Nov. 2019
Portland, ME